

Gerald R Ford (CVN 78) class (CVNM)

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Publication: Fighting Ships

UPDATED

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2023-03-09

Fleetlist

Ship totals				Gerald R Ford		
IN SERVICE				1		
BUILDING				2		
PROPOSED/TRANSFER				2		
<i>Name</i>	<i>No</i>	<i>Builders</i>	<i>Laid down</i>	<i>Launched</i>	<i>Commissioned</i>	<i>F/S</i>
Gerald R Ford	CVN 78	Huntington Ingalls Industries (Newport News Shipbuilding)	14 Nov 2009	17 Nov 2013	22 Jul 2017	AA
John F Kennedy	CVN 79	Huntington Ingalls Industries (Newport News Shipbuilding)	22 Aug 2015	16 Dec 2019	2024	Bldg
Enterprise	CVN 80	Huntington Ingalls Industries (Newport News Shipbuilding)	5 Apr 2022	2025	2028	Bldg

Doris Miller	CVN 81	Huntington Ingalls Industries (Newport News Shipbuilding)	—	—	—	Bldg
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2023-03-09

Specifications

Gerald R Ford	
Dimensions and weights	
Displacement	
full load:	101,605 tonnes (est) (100,000.3 (uk) t est) (112,000.3 t (short) est) (101,605,000 kg est)
Length	
overall:	332.8 m (1091.9 ft)
Beam	
overall:	78.0 m (255.9 ft)
waterline:	40.8 m (133.9 ft)
Draught	
hull:	12.4 m (40.7 ft)
Flight deck	
length overall:	332.8 m (1091.9 ft)
width overall:	78.0 m (255.9 ft)
launch mechanism:	catapult
Aircraft lift	
number:	3

Performance	
Speed	
top speed:	30 kt (est) (55.6 km/h est) (34.5 mph est)
Capacity	
Complement	
military	
crew:	4,297
officers:	508
Machinery:	Nuclear; 2 Bechtel A1B reactors; 4 shafts; fp props
Firepower	
Missiles:	SAM: 2 Raytheon Mk 29 GMLS [Ref 1] octuple launchers for Raytheon RIM-162D ESSM Block I; inertial guidance with mid-course update and semi-active radar terminal homing to 55 km (29.7 n miles) at Mach 3.6; warhead 39 kg, or ESSM Block II (from CVN 79). 2 Raytheon Mk 49 21-cell launchers; 42 Raytheon RIM-116B RAM Block I, or Block 2A/2B (from CVN 79) [Ref 2].
Guns:	3 Raytheon Mk 15 Phalanx Block 1B with M61A1 20 mm/99; 6 barrels per mount; 4,500 rds/min combined to 1.5 km (0.8 n miles); on-mount search radar and EO/IR director [Ref 3]. 4 BAE Systems Mk 38 Mod 3 with M242 25 mm/87 Bushmaster. 12.7 mm MGs.
Physical countermeasures:	Decoys: SLQ-25 NIXIE.
Electronic countermeasures:	ESM/ECM: SLQ-32(V)6 SEWIP Block 2. SRQ-6 SSEE.
Radars:	Air search: CVN 78: Dual Band Radar (DBR) comprised of Raytheon SPY-3 [Ref 4]; I-band and Lockheed Martin SPY-4 Volume Search Radar (VSR) [Ref 5]; E/F-band. CVN 79: Raytheon SPY-6(V)3 Enterprise Air Surveillance Radar; E/F-band. Air/surface search: CVN 79: Raytheon SPQ-9B; I-band. Navigation: SPS-73V(12); I-band.

	Fire control: CVN 78: SPY-3; I-band. From CVN 79: 4 Mk 9; target illumination; I-band (2 per Mk 29 launcher). PAR: SPN-46. Tacan: URN-25. IFF: UPX-29(V).
Combat data systems:	SSDS Mk II (Baseline 12 from CVN 79). USG-2 Cooperative Engagement Capability (CEC) (CVN 78). CEC Block II from CVN 79. Links 4, 11, 16, 22.
Weapon control systems:	SSDS Mk II.
Fixed-wing aircraft:	Composition will depend on mission but will comprise 75+ aircraft (F-35C, F/A-18E/F, EA-18G, E-2D, CMV-22B, MQ-25A).
Helicopters:	MH-60R/S

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Programmes

Programmes:	Newport News Shipbuilding awarded a construction preparation contract in May 2004 for detailed design, component development, long-lead procurement, and advanced construction of the lead ship CVN 78. First steel cut on 11 August 2005. Detailed design and construction contract for CVN 78 signed in September 2008. First of class problems delayed commissioning of CVN 78 until 2017. A more efficient build strategy is being applied to CVN 79 whose primary construction contract was delayed until 5 June 2015 in order to implement CVN 78 lessons learned. Switch to single-phase delivery for CVN 79 instead of initially contracted dual-phase approach was confirmed in March 2020. A contract for advanced procurement of CVN 80 was awarded on 23 May 2016. To reduce acquisition costs, the navy awarded a block-buy contract modification on 31 January 2019 for the detail design and construction of two Ford-class aircraft carriers: CVN 80 and CVN 81. CVN 79 schedule changed in November 2020 to an earlier single-phase complete-ship delivery, slated for June 2024. CVN 82 is to be awarded in FY 2028.
Structure:	The Ford class flight deck and below deck have been optimised to increase sortie rates and weapons movement. This is to be accomplished with a new design and relocation of the island, three aircraft lifts, and advanced weapons elevators (AWE). Other features include four Electromagnetic Aircraft Launch Systems (EMALS), Advanced Arresting Gear (AAG) system, new Ship Self-Defense

System (SSDS) open architecture combat system, fully integrated warfare system, a new nuclear power plant, and a flexible ship architecture to support the rapid insertion of future warfighting technologies. Significant habitability improvements are to be incorporated. DBR (SPY-3 and SPY-4) is replaced by Enterprise Surveillance Suite including SPY-6(V3) EASR from CVN 79.

Operational: Planned service life 50 years with one mid-life refuelling. CVN 78 sailed for initial sea trials on 8 April 2017 and conducted a successful launch (by EMALS) and recovery (by AAG) of an F/A-18F on 28 July 2017. CVN 78 completed Flight Deck Certification and Carrier Air Traffic Control Center on 20 March 2020. Combat System Ship Qualification Trials completed 17 April 2021, followed by full shock trial. CVN 78 reached initial operational capability on 22 December 2021 and is based at Norfolk, Virginia. First official operational deployment took place 4 October 2022 through 26 November 2022.



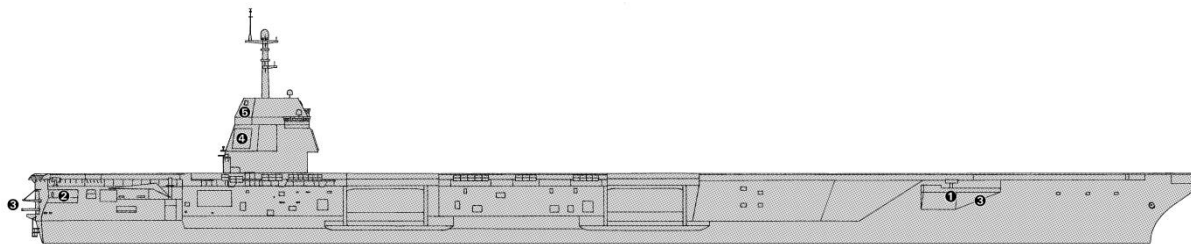
GERALD R FORD (with Nimitz-class carrier) (US Navy)

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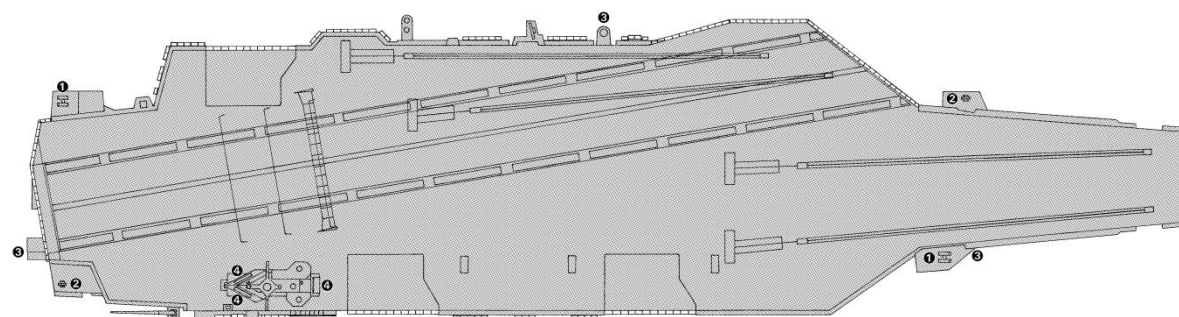
GERALD R FORD (US Navy)

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GERALD R FORD (Ian Sturton)

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GERALD R FORD (Ian Sturton)

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2023-07-14

Additional imagery

Online only images



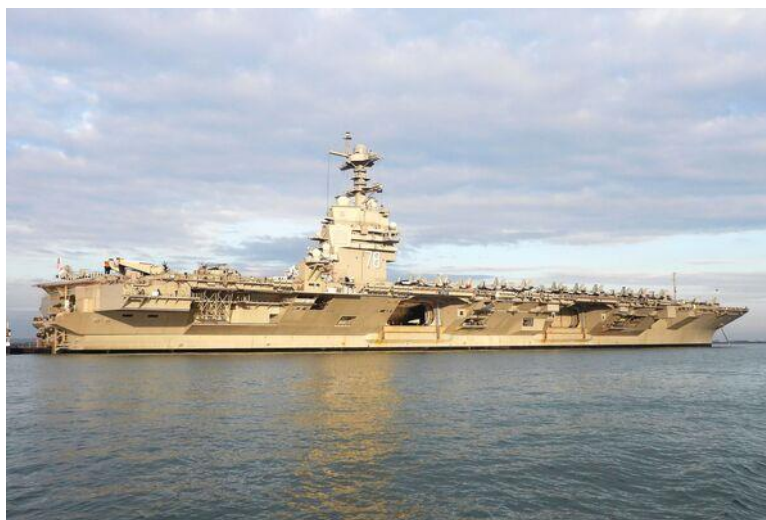
GERALD R FORD (M Declerck)

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GERALD R FORD (US Navy)

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GERALD R FORD (G&M Gyssels)

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GERALD R FORD (US Navy)

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GERALD R FORD (G&M Gyssels)

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GERALD R FORD CLASS SILHOUETTE (Ian Sturton)

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